# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 8 — SCIENCE TWO-CHAPTER TEST SET D

Chapters 7 & 8: **Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** Which sample has particles arranged in a regular close-packed way? 1

**(a)** oxygen **(b)** iron nail **(c)** water **(d)** steam

**2.** Solids do not mix easily because their particles: 1

**(a)** are far apart **(b)** are tightly held in place
**(c)** are free to move **(d)** have no forces

**3.** When water boils, it changes into: 1

**(a)** water vapour (gas) **(b)** salt
**(c)** ice **(d)** solid

**4.** The state of matter with fixed volume but no fixed shape is: 1

**(a)** solid **(b)** gas **(c)** all **(d)** liquid

**5.** The boiling point of water at atmospheric pressure is: 1

**(a)** 133 °C **(b)** 100 °C **(c)** 0 °C **(d)** 1538 °C

**6.** The black mass iron sulfide is not attracted by a magnet because it is a: 1

**(a)** mixture still **(b)** alloy
**(c)** element **(d)** new substance with different properties

**7.** Minerals are: 1

**(a)** natural solid substances with fixed chemical composition
**(b)** mixtures of random solids
**(c)** gases
**(d)** man-made liquids

**8.** Rust is a: 1

**(a)** pure element **(b)** alloy of iron
**(c)** compound (iron oxide) **(d)** mixture of gases

**9.** Which is a pure substance? 1

**(a)** Sea water **(b)** Distilled water **(c)** Milk **(d)** Soil

**10.** Compounds made only of non-metals include: 1

**(a)** iron and gold **(b)** salt and rust only
**(c)** brass and bronze **(d)** carbon dioxide and water

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 10–15 words each.*

**11.** Is ice an exception? Explain. 2

**12.** Why do we not see salt in sea water? 2

**13.** Name the gas formed when iron reacts with dilute hydrochloric acid and write the equation. 2

**14.** State two differences between mixtures and compounds. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 30–40 words each.*

**15.** Why does a solid expand on heating? Explain with particle ideas. 3

**16.** What would happen in daily life if water were a mixture of hydrogen and oxygen? 3

**17.** Explain how a drop of ink diffuses in water. 3

OR

Describe what would happen to a chair if all its constituent particles were removed. 3

**18.** Give three examples of pure substances from the kitchen. 3

OR

Differentiate between metals, non-metals and metalloids with one example each. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 80–100 words each.*

**19.** Describe three activities to show that particles of matter are very small. 5

OR

Prepare a concept note on states of matter linking particles, forces, shape, volume and temperature changes. 5

**20.** Explain whether air is a pure substance. Name five gases in it and suggest a way to show it is a mixture. 5

OR

Describe how chemists and metallurgists make use of mixtures and compounds with examples from alloys and minerals. 5

## END OF QUESTION PAPER — Total: 40 Marks

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